

PRANAV KISHAN T Y

+91 6382891980 | ty.pranavkishan7905@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Computer Science undergraduate experienced in secure backend engineering, full-stack web development, and cloud deployments. Proficient in Python, Java, and SQL with a strong foundation in zero-knowledge principles, modern DevOps practices, and responsible AI integration.

EDUCATION

Bachelor of Technology - Computer Science and Engineering	2023-2027
Amrita Vishwa Vidyapeetham - Coimbatore, India	CGPA - 8.66
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Computer Networks, Database Systems, Software Engineering, Machine Learning, Neural Networks and Deep Learning	

TECHNICAL SKILLS

- Languages:** Python, SQL, Java, C/C++, JavaScript, TypeScript
- AI & Emerging Tech:** Generative AI Fundamentals, AI Assistant Workflows
- Tools & Platforms:** Git, GitHub Actions, Docker, Linux, Bash
- Security:** AES/RSA Encryption, Digital Signatures, RBAC, Zero-Knowledge Principles
- Databases & Cloud:** PostgreSQL, MySQL, MongoDB, SQLite, Google Cloud Platform
- Backend & Systems:** FastAPI, Node.js, Express.js, REST APIs

PROJECTS

- InfoStuffs** | [Repo](#) | [Live](#) | [Blog](#) | React.js, MongoDB, Express.js, Node.js, Docker, Vercel
Zero-Knowledge personal information manager ensuring end-to-end privacy via client-side cryptography.
 - Architected a **zero-knowledge security model** using **client-side AES-256**, **Web Crypto API**, and **PBKDF2** to guarantee absolute **data confidentiality**.
 - Engineered a backend infrastructure unifying a persistent **Node.js Docker container** with stateless **Vercel Functions** for **cost-optimized scaling**.
 - Implemented **offline-first functionality** utilizing secure **client-side state caching**, ensuring high data availability and seamless synchronization without compromising the zero-knowledge encryption model.
- Netwatchpy** | [Repo](#) | [PyPI](#) | Python, SQLite, GitHub Actions
A cross-platform, installable TUI for real-time network usage monitoring.
 - Refactored **SQLite schema** from text strings to **Unix Epoch integers**, reducing **range query latency by 40%** via **optimized B-Tree indexing** to demonstrate **scalable data handling**.
 - Engineered a **fault-tolerant SQLite WAL architecture** to process **real-time telemetry** and render 24-hour bandwidth metrics **without database locks**.
 - Established a GitHub Actions CI/CD pipeline to automate cross-platform binary builds and PyPI releases, accumulating over 11,000 total downloads.
- SurplusSync** | [Repo](#) | [Live](#) | NestJS, React.js, PostgreSQL, Docker, GitHub Actions, Redis, Socket.io
Real-Time Food Redistribution Platform.
 - Directed **Agile workflows** as **Scrum Master**, managing **cross-functional team sprints** to ensure project deliverables conformed to **business requirements**.
 - Set up a **continuous integration pipeline** using **GitHub Actions** to automate testing and build integrity for a **containerized monorepo**.
 - Implemented **real-time event infrastructure** using **Socket.io** for live alerts and integrated **Redis for caching**, alongside automated **PostgreSQL database backups**.

EXPERIENCE

- Backend Developer - Magilam Foods (Client Project)** February 2026 - Present
 - Evaluated client requirements to design and develop an inventory management backend, ensuring **strict database consistency** across complex order workflows.
- Intel IoT Club - Full-Stack Developer** November 2025 - Present
 - Resolved critical **server-client rendering mismatches** (hydration errors) to ensure **visual stability** on the official platform.
 - Implemented **modular UI components** for the EasyEPFO payment interface, improving core **visual consistency**.

ACHIEVEMENTS

- Code with Cisco 2026**, Finalist, recognized for outstanding problem-solving and programming skills.
- Built an **automated web research MCP** server that equips AI agents with **real-time scraping capabilities**, securing a top-40 leaderboard position among 25,000+ competitors in the Puch.AI Hackathon.
- Advancing to Phase 2 of the Visteon Scholar Program**, A competitive mentorship initiative for engineering UGs.

CERTIFICATIONS AND COURSES

- IETE Docker Workshop | [CERTIFICATE](#)
- Google Cloud - Introduction to GenAI | [CERTIFICATE](#)
- Google Cloud - Prompt Design in Vertex AI | [CERTIFICATE](#)
- Introduction to Distributed SQL and CockroachDB, Cockroach University | [CERTIFICATE](#)